MILITARY SPECIFICATION

TEST SET, ARMAMENT WIRING AN/AWM-38

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The test equipment covered by this specification shall be for flight line use and shall be used to conduct insulation and continuity tests of all Navy Aircraft Monitor and Control (AMAC) systems.

1.2 Classification - The test equipment covered by this specification shall consist of the items listed below and the items listed in 1.3. The items listed below shall be considered as the basic components of the test set. The items listed in 1.3 shall not be considered as basic components of the test set, but they are required as indicated in table VIII and 3.5.6 for testing the AMAC systems.

Item	Type Designation or Drawing Number	Applicable Paragraph
Test Set, Armament Wiring	TS-2678/AWM-38	3.5.1
Programming Module Assembly	67A47D20-1	3.5.2
Programming Module Assembly	67A47D20-2	3.5.2
Case, Test Set	CY-6475/AWM-38	3.5.3
Cable, Power CA-1	57A116C57	3.5.4
Cable, Extension CA-2	57A116D23	3.5.5

1-3 Accessory Equipment - The test equipment when used in conjunction with AMAC systems shall operate with the following accessory cables:

FSC -1190

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Item	Drawing Number	Applicable Paragraph
Cable, Special Adapter CA-56	57A116F302	3.5.6.1
Cable, Power Conversion CA-27	57A116C75	3.5.6.2
Shorting Cable CA-47	57A116C131	3.5.6.3
Shorting Cable CA-25	57A116D73	3.5.6.3
Shorting Cable CA-32	57A116C80	3.5.6.3
Shorting Cable CA-30	57A116C78	3.5.6.3
Shorting Cable CA-42	57A116C90	3.5.6.3
Adapter Cable CA-10	57A116D21	3.5.6.3
Adapter Cable CA-54	57A116F274	3.5.6.3
Adapter Cable CA-21	57A116D58	3.5.6.3
Adapter Cable CA-28	57A116D76	3.5.6.3

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation of Delivery for
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

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MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS17322-1	Meter, Time Totalizing, 115 Volt, 400-Cycle, Rev. B

AERONAUTICAL BULLETIN

Air Force-Navy

ANA BULLETIN 400	Electronic Equipment; Aircraft and Guided Missiles, Applicable Documents

DRAWINGS

Naval Air Systems Command

DL 67A47F1	Data List for Test Set, Armament Wiring AN/AWM-38 and all drawings therein

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials -

3.2.1 General Requirements - The parts and materials specified in the drawings and related documents for use in construction of this equipment were selected to ensure the fulfillment of all requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change, and the cost associated therewith, upon the contractor. Approval for deviations must be requested from the procuring activity by the contractor and granted by the procuring activity prior to submission of equipments for preproduction tests. A request for approval shall include, as supporting evidence, the data specified in subparagraphs of 3.2.2 (Supplemental Requirements). Approval of parts and material deviations shall not be construed by the contractor to constitute a waiver of the performance requirements or any other requirements of this specification.

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3.2.2 Supplemental Requirements - In the event that the contractor proposes to alter the equipment design delineated by this specification, these supplemental requirements for the selection of parts and materials are applicable.

3.2.2.1 Basic Conformance - Parts and materials selected for use in the equipment shall conform to the requirements of MIL-T-21200. Parts shall be selected from those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform adequately in the intended applications and environments specified herein.

3.2.2.2 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained by the contractor as outlined in the basic conformance specification of 3.2.2.1.

3.3 Design and Construction - The test set shall be manufactured in accordance with the drawings and subsidiary specifications listed in Data List DL 67A47F1 and related lists, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this and related specifications. In any case it is a condition of final acceptance that the equipment manufactured meet all the performance requirements of this specification, and such acceptance test documents as may be applicable. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be specifically approved by the procuring activity prior to their incorporation.

3.3.1 Errors and/or Omissions - Previous design review, prototype development, and testing has established a high confidence level in the adequacy of this design and associated drawings on DL 67A47F1.

3.3.1.1 Patent Errors - Prior to the release of any drawing by the contractor for procurement or fabrication, each drawing is to be reviewed by appropriate and competent technical personnal to identify any PATENT (apparent) errors or omissions. It is the noted responsiblity of the contractor to insure that this review is accomplished and that errors or omissions are promptly brought to the procuring contract officer for resolution. Resolution of all problems associated with any drawing must be completed before that drawing is released by the contractor for procurement or manufacture.

3.3.1.2 Latent Errors - Latent (hidden) errors or omissions discovered during procurement or fabrication cycles which result in a part, component, or assembly not meeting its design requirements, specification test requirements or design intent shall also be promptly brought to the attention of the procuring contracts officer for resolution.

3.3.1.3 Workmanship Defects - The contractor is also responsible for assuring that discrepancies existing between hardware manufactured to the drawings and tested to the specification are not the result of workmanship defects.

3.3.1.4 Contractor Negotiation - Official negotiation will determine the need for a waiver, authorized deviation, or design change and establish due compensation to the contractor.

3.3.1.5 Total Weight - The total weight of the equipment, including cables, shall not exceed 76.5 pounds.

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3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 50 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 200 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The test set shall contain the following time totalizing meter in accordance with MIL-M-7793.

Unit	Type of Meter
Test Set	MS17322-1

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200.

3.3.3.2 Cable Assemblies, Power Electrical - Applicable input power cable and connections shall conform to 57A116C57.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-T-21200. The drawings listed in Data List DL 67A47F1 shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-STD-461, Class ID, except as listed as follows:

Test Code	Extended Limits
CE 01	20 DB
CE 02	20 DB
CE 03	10 DB
CE 04	10 DB
CS 01	Refer to 3.3.5.1
RE 02	10 DB

3.3.5.1 Susceptibility Test CS 01 Exception - HI-POT lamp flickering "ON" and "OFF" at approximately 400 and 800 Hz during the conducted susceptibility test, CS 01, shall not be cause for rejection.

3.3.6. Maintenance Provisions and Field Testing - Provisions for maintenance

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shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground (10,000 ft max.)
Vibration	Not Applicable
Shock	Not Applicable
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.3 VAC, 400 ± 5 Hz

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment, except as modified herein:

3.3.9.1 Explosion Proof - Explosion proof testing shall not be required.

3.3.9.2 Salt Spray - Operation of the test set during salt spray tests shall not be required. Upon completion of the tests, the test set shall be removed from the instrument case, side panels removed and internally washed with tap water and dried for 16 hours at 50°C prior to operation.

3.3.9.3 Sand and Dust - Sand and dust tests shall be performed at +55° Centigrade.

3.3.9.4 Fungus - Fungus resistance tests shall not be performed on the test set.

3.3.10 Warm-Up Time - No warm-up time shall be required.

3.3.11 Primary Input Power Requirements - The test equipment shall be capable of satisfactory operation when supplied normal input power having characteristics as defined in MIL-STD-704, Cat B, and as modified herein. The power required shall be of the following type and shall not exceed the specified amount.

AC Power, Single Phase, 115 ± 10V, 400 Hz, 100 VA

3.3.11.1 Protection - The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in MIL-STD-704, Cat C.

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3.4 Performance - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

3.5 Detail Requirements -

3.5.1 Test Set, Armament Wiring TS-2678/AWM-38 - The test set shall be manufactured in accordance with the documents listed on DL 67A47F1 and shall meet the following requirements:

3.5.1.1 Function - The test set shall be used as a power supply and circuit selector when conducting high potential (hi-pot) tests and as a continuity tester on the following AMAC systems:

Aero 6B	T-375
A/A24B-4(V)	T-414-101
T-249A	T-414-57

3.5.1.2 Weight - The weight of the test set including the bottom case shall not exceed 43 pounds.

3.5.1.3 Contents - The test set shall contain the following:

(1) Hi-Pot Circuits

(2) Continuity Circuits

(3) Check Switch Circuit

(4) System Lamp Circuit

(5) Electromagnetic Radiation Filters

3.5.1.3.1 Hi-Pot Lamp Circuit - The hi-pot lamp circuit shall meet the following requirements:

3.5.1.3.1.1 Hi-Pot Circuit Adjustment - The HI-POT lamp circuit shall be adjusted at standard conditions to conform to the following requirements when inputs and switch positions are as specified.

(1) With the specified input voltage applied to connector 1A1A1J1, place the AC PWR ON-OFF switch to the ON position and the HI-POT LAMP (OFF-ON) selector to the HI-POT LAMP ON position, the HI-POT lamp shall light.

(2) Place the HI-POT LAMP (OFF-ON) selector to the HI-POT LAMP OFF position, the HI-POT lamp shall extinguish.

(3) If the HI-POT lamp does not operate as indicated in 3.5.1.3.1.1(1) and (2), place the HI-POT LAMP (OFF-ON) selector to the HI-POT

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LAMP ON position and if necessary, adjust potentiometer 1A2R5 clockwise so that the HI-POT lamp shall light without blinking. Place the HI-POT LAMP (OFF-ON) selector to the HI-POT LAMP OFF position, if necessary, adjust 1A2R5 counterclockwise until the lamp extinguishes. Turn the HI-POT LAMP (OFF-ON) selector ON and OFF and observe that the lamp shall light with the selector in the HI-POT LAMP ON position.

3.5.1.3.1.2 Hi-Pot Circuit Voltages - After the requirements of 3.5.1.3.1.1 have been met, hi-pot circuit voltages shall be as specified in tables I through IV. Inputs, loads, and switch positions shall be as specified in tables I through IV. The hi-pot circuit voltages shall be within tolerance for any combination of input voltage and frequency within the limits specified in the column heading. Measurements shall be made with meters that have the minimum sensitivities and accuracies specified in tables I through IV.

3.5.1.3.1.3 Hi-Pot Voltage Outputs - The hi-pot voltage output shall be 615 ± 40 volts d-c when measured between pin 6 of 1A1A1J2(-) and each of the pins of 1A1A1J2(+) as listed in table V. Inputs and switch positions shall be as specified in table V with the master module on programming module assembly (67A47D20-2) in the test set circuit. The voltage measurements shall be taken with a voltmeter that has an input impedance of 10 megohms or greater and an accuracy of ±2 percent. Front panel system lamps 1A1DS5 through 1A1DS10 shall be lit when making the voltage measurements.

3.5.1.3.1.4 Hi-Pot Lamp Circuit - With the AC PWR ON-OFF switch in the OFF position, the HI-POT selector in position 17, the FUNCTION switch set for HI-POT and the master module on programming module assembly (67A47D20-2) in the test set circuit, the HI-POT lamp shall light when 28 ± 2 volts a-c or d-c is applied to pins n and m of 1A1A1J2.

3.5.1.3.1.5 Hi-Pot Circuit Continuity - With the HI-POT lamp bulb removed, the AC PWR ON-OFF switch in the OFF position, the HI-POT selector in position 17, the FUNCTION switch set for HI-POT and the master module on programming module assembly (67A47D20-2) in the test set circuit, the resistance between pin n of 1A1A1J2 and each of the following pins of 1A1A1J2 shall be 100 ± 17 ohms: A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, W, X, Z, a, b, c, d, e, f, g, h, j, k, and m. (Measurements shall be made with a 5 percent minimum accuracy ohmmeter.)

3.5.1.3.1.6 Hi-Pot Loading - Hi-pot loading shall meet the following requirements when measured with a ± 0.3 percent minimum accuracy ohmmeter:

(1) With the AC PWR ON-OFF switch in the OFF position and the HI-POT LAMP (OFF-ON) switch in the "NORMAL" position, the resistance between pins 17 and 18 of connector 1A2J1 shall be 116 ± 1.16K ohms.

(2) With the AC PWR ON-OFF switch in the OFF position and the HI-POT LAMP (OFF-ON) switch in the "NORMAL" position, the resistance between pins 15 and 16 of connector 1A2J1 shall be 124 ± 1.24K ohms.

3.5.1.3.1.7 Hi-Pot Circuit Performance - After the requirements of 3.5.1.3.1.1 have been met, observe that the front panel HI-POT lamp shall light, under both standard and service conditions, when the test set is cycled in steps (1) through (7) as follows:

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Table I. Hi-Pot Circuit Voltages - Low Load (116 ± 1.16K ohms)

(Standard Conditions)

Measure Between Test Points or Terminals			Voltmeter		Standard Conditions HI-POT LAMP switch to LAMP ON position	
D-C Meter Leads (when required)		Type Test	Meter Sensitivity (ohms/volt-min.)	Meter Accuracy (percent)	Input 115 ± 1.3 volts 400 ± 5 Hz	Input 115 ± 10 volts 400 ± 20 Hz
+	-					
1A2TP5	1A2TP6	A-C	1 K	±0.1	115 ± 1.3 v a-c	115 ± 10 v a-c
1A2TP5	1A2TP6	Freq.	1 K	±0.3	400 ± 5 Hz	400 ± 20 Hz
1A2TP2	1A2TP3	D-C	20 K	±2.0	500 ± 30 v d-c	500 ± 80 v d-c
1A2TP2	1A2TP3	Ripple	20 K	±3.0	5.0 v a-c (max.)	5.0 v a-c (max.)
1A2TP1	1A2TP3	A-C	20 K	±3.0	455 ± 25 v a-c	455 ± 65 v a-c
1A2Q1-2	1A2TP3	D-C	50 K	±2.0	13.0 ± 1.5 v d-c	13.0 ± 1.5 v d-c
1A2Q1-1	1A2TP3	D-C	50 K	±2.0	12.5 ± 1.5 v d-c	12.5 ± 1.5 v d-c
1A2Q1-3	1A2TP3	D-C	50 K	±2.0	27.0 ± 2.5 v d-c	27.0 ± 6 v d-c
1A2Q1-2	1A2Q1-1	D-C	50 K	±2.0	0.55 ± 0.15 v d-c	0.55 ± 0.15 v d-c
1A2TP7	1A2TP8	A-C	1 K	±3.0	28.0 ± 2.0 v a-c	28.5 ± 4.5 v a-c
1A2T1-7	1A2T1-8	A-C	1 K	±3.0	30.0 ± 3.0 v a-c	30.0 ± 5.5 v a-c
1A2T1-9	1A2T1-8	A-C	1 K	±3.0	30.0 ± 3.0 v a-c	30.0 ± 5.5 v a-c
1A2TP4	1A2TP3	D-C	20 K	±2.0	26.0 ± 2.5 v d-c	26.5 ± 5.5 v d-c
1A2TP9	1A2TP3	A-C	20 K	±3.0	42.5 ± 4.5 v a-c	43.0 ± 7.5 v a-c

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Table II. Hi-Pot Circuit Voltages - Low Load (116 ± 1.16K ohms)

(Service Conditions)

Measure Between Test Points or Terminals			Voltmeter		Service Conditions HI-POT LAMP switch to LAMP ON position	
D-C Meter Leads (when required)		Type Test	Meter Sensitivity (ohms/volt-min.)	Meter Accuracy (percent)	Input 115 ± 1.3 volts 400 ± 5 Hz	Input 115 ± 10 volts 400 ± 20 Hz
+	-					
1A2TP5	1A2TP6	A-C	1 K	±0.1	115 ± 1.3 v a-c	115 v ± 10 v a-c
1A2TP5	1A2TP6	Freq.	1 K	±0.3	400 ± 5 Hz	400 ± 20 Hz
1A2TP2	1A2TP3	D-C	20 K	±2.0	500 ± 30 v d-c	500 ± 80 v d-c
1A2TP2	1A2TP3	Ripple	20 K	±3.0	5.0 v a-c (max.)	5.0 v a-c (max.)
1A2TP1	1A2TP3	A-C	20 K	±3.0	455 ± 25 v a-c	455 ± 65 v a-c
1A2Q1-2	1A2TP3	D-C	50 K	±2.0	13.0 ± 2.0 v d-c	13.0 ± 2.0 v d-c
1A2Q1-1	1A2TP3	D-C	50 K	±2.0	12.5 ± 2.5 v d-c	12.5 ± 2.5 v d-c
1A2Q1-3	1A2TP3	D-C	50 K	±2.0	27.0 ± 2.5 v d-c	27.0 ± 6 v d-c
1A2Q1-2	1A2Q1-1	D-C	50 K	±2.0	0.55 ± 0.30 v d-c	0.55 ± 0.30 v d-c
1A2TP7	1A2TP8	A-C	1 K	±3.0	28.0 ± 2.0 v a-c	28.5 ± 4.5 v a-c
1A2T1-7	1A2T1-8	A-C	1 K	±3.0	30.0 ± 3.0 v a-c	30.0 ± 5.5 v a-c
1A2T1-9	1A2T1-8	A-C	1 K	±3.0	30.0 ± 3.0 v a-c	30.0 ± 5.5 v a-c
1A2TP4	1A2TP3	D-C	20 K	±2.0	26.0 ± 2.5 v d-c	26.0 ± 5.5 v d-c
1A2TP9	1A2TP3	A-C	20 K	±3.0	42.5 ± 4.5 v a-c	43.0 ± 7.5 v a-c

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Table III. Hi-Pot Circuit Voltages - High Load (124 ± 1.24K ohms)

(Standard Conditions)

Measure Between Test Points or Terminals			Voltmeter		Standard Conditions HI-POT LAMP switch to LAMP OFF position	
D-C Meter Leads (when required)		Type Test	Meter Sensitivity (ohms/volt-min.)	Meter Accuracy (percent)	Input 115 ± 1.3 volts 400 ± 5 Hz	Input 115 ± 10 volts 400 ± 20 Hz
+	-					
1A2TP5	1A2TP6	A-C	1 K	±0.1	115 ± 1.3 v a-c	115 ± 10 v a-c
1A2TP5	1A2TP6	Freq.	1 K	±0.3	400 ± 5 Hz	400 ± 20 Hz
1A2TP2	1A2TP3	D-C	20 K	±2.0	503 ± 30 v d-c	503 ± 80 v d-c
1A2TP2	1A2TP3	Ripple	20 K	±3.0	5.0 v a-c (max.)	5.0 v a-c (max.)
1A2TP1	1A2TP3	A-C	20 K	±3.0	455 ± 25 v a-c	455 ± 65 v a-c
1A2Q1-2	1A2TP3	D-C	50 K	±2.0	13.7 v d-c (max.)	13.7 v d-c (max.)
1A2Q1-1	1A2TP3	D-C	50 K	±2.0	13.0 v d-c (max.)	13.0 v d-c (max.)
1A2Q1-3	1A2TP3	D-C	50 K	±2.0	28 ± 2.5 v d-c	28 ± 5 v d-c
1A2Q1-2	1A2Q1-1	D-C	50 K	±2.0	0.7 v d-c (max.)	0.7 v d-c (max.)
1A2TP7	1A2TP8	A-C	1 K	±3.0	28.0 ± 2 v a-c	28.0 ± 4.5 v a-c
1A2T1-7	1A2T1-8	A-C	1 K	±3.0	31.0 ± 2 v a-c	31.0 ± 4 v a-c
1A2T1-9	1A2T1-8	A-C	1 K	±3.0	31.0 ± 2 v a-c	31.0 ± 4 v a-c
1A2TP4	1A2TP3	D-C	20 K	±2.0	27.0 ± 2 v d-c	27.5 ± 4.5 v d-c
1A2TP9	1A2TP3	A-C	20 K	±3.0	42.5 ± 4.5 v a-c	43.0 ± 7.5 v a-c

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Table IV. Hi-Pot Circuit Voltages - High Load (124 ± 1.24K ohms)

(Service Conditions)

Measure Between Test Points or Terminals			Voltmeter		Service Conditions HI-POT LAMP switch to LAMP OFF position	
D-C Meter Leads (when required)		Type Test	Meter Sensitivity (ohms/volt-min.)	Meter Accuracy (percent)	Input 115 ± 1.3 volts 400 ± 5 Hz	Input 115 ± 10 volts 400 ± 20 Hz
+	-					
1A2TP5	1A2TP6	A-C	1 K	±0.1	115 ± 1.3 v a-c	115 ± 10 v a-c
1A2TP5	1A2TP6	Freq.	1 K	±0.3	400 ± 5 Hz	400 ± 20 Hz
1A2TP2	1A2TP3	D-C	20 K	±2.0	503 ± 30 v d-c	503 ± 80 v d-c
1A2TP2	1A2TP3	Ripple	20 K	±3.0	5.0 v a-c (max.)	5.0 v a-c (max.)
1A2TP1	1A2TP3	A-C	20 K	±3.0	455 ± 25 v a-c	455 ± 65 v a-c
1A2Q1-2	1A2TP3	D-C	50 K	±2.0	14.1 v d-c (max.)	14.1 v d-c (max.)
1A2Q1-1	1A2TP3	D-C	50 K	±2.0	13.3 v d-c (max.)	13.3 v d-c (max.)
1A2Q1-3	1A2TP3	D-C	50 K	±2.0	28 ± 2.5 v d-c	28 ± 5 v d-c
1A2Q1-2	1A2Q1-1	D-C	50 K	±2.0	0.85 v d-c (max.)	0.85 v d-c (max.)
1A2TP7	1A2TP8	A-C	1 K	±3.0	28.0 ± 2 v a-c	28.0 ± 4.5 v a-c
1A2T1-7	1A2T1-8	A-C	1 K	±3.0	31.0 ± 2 v a-c	31.0 ± 4 v a-c
1A2T1-9	1A2T1-8	A-C	1 K	±3.0	31.0 ± 2 v a-c	31.0 ± 4 v a-c
1A2TP4	1A2TP3	D-C	20 K	±2.0	27.0 ± 2 v d-c	27.5 ± 4.5 v d-c
1A2TP9	1A2TP3	A-C	20 K	±3.0	42.5 ± 4.5 v a-c	43.0 ± 7.5 v a-c

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Table V. Hi-Pot Voltage Output

Input: 115.0 ± 1.3 volts, 400 ± 5 Hz applied to pins C and D of 1A1A1J1.

AC PWR ON-OFF: ON position

FUNCTION Switch: Set for hi-pot function.

Indication: 615 ± 40 volts d-c.

Meter Connections: Pin 6 of 1A1A1J2(-) and following pins 1A1A1J2(+).

PROGRAM Switch in Position A		PROGRAM Switch in Position B	
HI-POT Selector Switch Position	1A1A1J2 pin	HI-POT Selector Switch Position	1A1A1J2 pin
1	A	1	T
2	B	2	U
3	C	3	V
4	D	4	W
5	E	5	X
6	F	6	Z
7	G	7	a
8	H	8	b
9	J	9	c
10	K	10	d
11	L	11	e
12	M	12	f
13	N	13	g
14	P	14	h
15	R	15	j
16	S	16	k
		16	p

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Table VI. Hi-Pot Selector Switch Position vs Pin Loading

PROGRAM Switch in Position A		PROGRAM Switch in Position B	
HI-POT Selector Switch Position	1A1A1J2 pin	HI-POT Selector Switch Position	1A1A1J2 pin
1	A	1	T
2	B	2	U
3	C	3	V
4	D	4	W
5	E	5	X
6	F	6	Z
7	G	7	a
8	H	8	b
9	J	9	c
10	K	10	d
11	L	11	e
12	M	12	f
13	N	13	g
14	P	14	h
15	R	15	j
16	S	16	k
		16	p

(1) Insert the master module on programming module assembly (67A47D20-2) into the test set circuit.

(2) Connect a jumper wire across pins δ and n connector 1A1A1J2.

(3) Set the FUNCTION switch for HI-POT.

(4) In conjunction with steps (5), (6), and (7), connect in turn, an external resistor of 101 ± 1 K ohms across shorted pins δ and n and each of the pins listed in table VI.

(5) Place the AC PWR ON-OFF switch to the ON position.

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(6) Place the HI-POT selector switch to the position in table VI corresponding to the pin that has the 101 ± 1 K ohm resistor connected to it.

(7) Place the PROGRAM switch to position A or B as required by the pin connection of table VI.

3.5.1.3.2 Continuity Circuits - With the AC PWR ON-OFF switch in the OFF position, the master module on programming module assembly (67A47D20-2) in the test circuit, the FUNCTION switch set for CONTINUITY, the resistance values between the pins of connector 1A1A1J2 listed in table VII shall be observed when the PROGRAM and CONTINUITY selector switches are placed in the specified positions. Minimum ohmmeter accuracy, used for the resistance measurements, shall be ± 25 percent over the specified resistance range.

3.5.1.3.3 Check Switch Circuit - The check switch circuit shall meet the following requirements:

3.5.1.3.3.1 Check Switch Lamp Indication - After the equipment meets the requirements specified in 3.5.1.3.1 and 3.5.1.3.2 and with the master module on the programming module assembly (67A47D20-2) in the test set circuit, pins n and δ of 1A1A1J2 connected together, the PROGRAM selector placed in position A, the HI-POT selector placed in the OFF position, the FUNCTION switch set for HI-POT, AC PWR OFF-ON switch in the ON position and the unit activated, the HI-POT and CONTINUITY lamps shall light while the CHECK switch is pressed. The HI-POT and CONTINUITY lamps shall extinguish when the CHECK switch is released.

3.5.1.3.3.2 Check Switch Loading - With the AC PWR ON-OFF switch in the OFF position, the observed resistance between pins 6 and 7 of connector 1A2J1 shall be 102 ± 1.0 K ohms when measured with a ± 0.3 percent minimum accuracy ohmmeter.

3.5.1.3.4 System Lamp Circuit - With standard input power applied to connector 1A1A1J1 and the AC PWR ON-OFF switch in the ON position, only the front panel system lamp designated A/A24B-4(V), T414-101, AERO 6B, T414-57, T249A or T375 shall light when the corresponding system program module is connected into the test set circuit.

3.5.1.3.5 Electromagnetic Radiation Filters - Electromagnetic radiation filters shall meet the following requirements:

3.5.1.3.5.1 Power Circuit Filters - The 115-volt, 400-Hz input power circuit shall contain two electromagnetic radiation filters that shall function with minimum insertion losses as specified in drawing 57A116C203-1.

3.5.1.3.5.2 Test Circuit Filters - Each of the thirty-seven circuits connected at the front panel TEST receptacle when used for hi-pot or contituity tests shall function with minimum insertion losses as specified in drawing 57A116C203-2.

3.5.1.4 Controls -

(1) AC PWR ON-OFF switch shall be a two position toggle switch used for turning the power source on and off.

(2) PROGRAM selector, A and B positions, shall be a two position rotary type switch used for selecting applicable decks of the HI-POT and CONTINUITY selectors.

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Table VII. Continuity Circuit Resistance Values

PROGRAM switch in Position A		
CONTINUITY selector switch position	Between Pins of 1A1A1J2	*Resistance (ohms)
1	A and R	16 or less
1	K and f	8 to 220
1	K and ó	8 to 220

PROGRAM switch in Position B		
CONTINUITY selector switch position	Between Pins of 1A1A1J2	Resistance (ohms)
1	R and K	8 to 220
2	L and S	8 to 220
3	g and E	8 to 220
4	h and G	8 to 220
5	j and J	8 to 220
6	k and M	8 to 220
7	m and N	8 to 220
8	Z and P	8 to 220
9	a and D	8 to 220
10	X and F	8 to 220
11	T and H	8 to 220
12	U and b	8 to 220
13	V and C	8 to 220
14	W and B	8 to 220
15	n and c	8 to 220
16	p and d	8 to 220
17	r and e	8 to 220

*NOTE: All resistance measurements, except the 16-ohm measurement across pins A and R, will be low in value initially and then will increase as the CONTINUITY lamp warms up.

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(3) HI-POT selector shall be an eighteen position rotary type switch used to select the circuits to be hi-pot tested and shall be labeled 1 through 17 and OFF.

(4) CONTINUITY selector shall be an eighteen position rotary type switch used to select circuits to be tested for continuity and shall be labeled 1 through 17 and OFF.

(5) CHECK switch shall be a spring-loaded press-type switch used to check the operation of the hi-pot and continuity circuits.

(6) FUNCTION switch shall be a two-position switch that shall program the test set for either hi-pot or continuity tests.

(7) HI-POT LAMP (OFF-ON) switch shall be a three-position, spring-loaded type switch used to calibrate the hi-pot circuit. When placed in the HI-POT LAMP ON position, an internal load of 116 ± 1.2 K ohms shall be applied to the hi-pot circuit and when placed in the HI-POT LAMP OFF position, an internal load of 124 ± 1.2 K ohms shall be applied to the hi-pot circuit. In the center position (no designation), the hi-pot circuit shall be connected to determine faults in the AMAC system.

3.5.1.5 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
1A1A1J1	MS3102A-18-10P	Power
1A1A1J2	MS3102A-28-21S	Test Circuit

3.5.2 Programming Module Assembly - The programming module assemblies 67A47D20-1 and 67A47D20-2 shall be manufactured in accordance with the documents listed on DL 67A47F1 and shall meet the following requirements:

3.5.2.1 Function - The programming module assemblies shall provide the individual modules required for hi-pot and continuity programming of the AMAC systems.

3.5.2.2 Weight - The weight of each programming module assembly shall not exceed 12 ounces.

3.5.2.3 Contents - The programming module assemblies shall contain individual hi-pot and continuity AMAC programming modules and a master module as follows:

Programming Module Assembly (67A47D20-1)	Programming Module Assembly (67A47D20-2)
A/A24B-4(V)	MASTER
T375	T414-101
T249A	T414-57
AERO 6B	

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3.5.3 Case, Test Set CY-6475/AWM-38 - The test set case shall be manufactured in accordance with the documents listed on DL 67A47F1 and shall meet the following requirements.

3.5.3.1 Function - The test set case shall be used to house the power cable CA-1, extension cable CA-2 and handbook NAVAIR 16-30AWM38-1.

3.5.3.2 Weight - The weight of test set case shall not exceed 17 pounds.

3.5.4 Cable, Power CA-1 - The power cable shall be manufactured in accordance with 57A116C57 and shall meet the following requirements.

3.5.4.1 Function - The power cable shall be used to connect the test set to a 115-volt, 400 Hz power source.

3.5.4.2 Weight - The weight of the power cable shall not exceed 3 pounds.

3.5.5 Cable, Extension CA-2 - The extension cable shall be manufactured in accordance with 57A116D23 and shall meet the following requirements:

3.5.5.1 Function - The extension cable plus the applicable accessory cable shall be used to connect the test set to the AMAC system under test.

3.5.5.2 Weight - The weight of the extension cable shall not exceed 12 pounds.

3.5.6 Accessory Cables - Accessory cables shall include special adapter cable CA-56, power conversion cable CA-27, and the cables listed in table VIII.

3.5.6.1 Cable, Special Adapter CA-56 - The special adapter cable shall be manufactured in accordance with 57A116F302 and shall meet the following requirements:

3.5.6.1.1 Function - The special adapter cable shall be used in place of the pylon on A-7A/B aircraft when testing.

3.5.6.2 Cable, Power Conversion CA-27 - The power conversion cable shall be manufactured in accordance with 57A116C75 and shall meet the following requirements:

3.5.6.2.1 Function - The power conversion cable shall be used to adapt connector P226 of power cable CA-1 to the power source connection used in S-2D/E aircraft.

3.5.6.3 Shorting and Adapter Cables - The shorting and adapter cables shall be manufactured in accordance with the applicable drawing listed in table VIII and shall meet the following requirements.

3.5.6.3.1 Function - Each accessory cable listed in table VIII of this specification, when used in conjunction with the test set, shall enable the test set to perform satisfactory tests as specified in the applicable aircraft special weapons system check list.

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Table VIII. Accessory Cable Chart

AMAC System	Shorting Cable		Adapter Cable	
	Cable Assembly	Drawing Number	Cable Assembly	Drawing Number
Aero 6B	----	-----	CA-10	57A116D21
A/A24B-4(V)	----	-----	CA-54	57A116F274
T-249A	CA-47 (See Note 1)	57A116C131	CA-21	57A116D58
T-375	CA-25 (See Note 2) CA-32 (See Note 3)	57A116D73 57A116C80	CA-21	57A116D58
T-414-101	CA-30 (See Note 4) CA-42 (See Note 4)	57A116C78 57A116C90	CA-28	57A116D76
T-414-57	CA-30 (See Note 5) CA-42 (See Note 5)	57A116C78 57A116C90	CA-21	57A116D58

NOTES:

1. Shorting cable CA-47 is used on the other T-249A weapon connector of the station being tested during continuity checks.
2. Shorting cable CA-25 is used on all other T-375 weapon stations during hi-pot tests.
3. Shorting cable CA-32 is used on the other T-375 weapon connector of the station being tested during continuity checks.
4. Shorting cable CA-30 is used on all other T-414-57 weapon stations and shorting cable CA-42 is used on all other T-414-101 weapon stations during hi-pot tests.
5. Shorting cable CA-30 is used on all other T-414-101 weapon stations and shorting cable CA-42 is used on all other T-414-57 weapon stations during hi-pot tests.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved tests procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature ratings and their applied circuit voltage, current, ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with MIL-T-21200.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall

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in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passsing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of 10 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 25 to 55 cps (excluding resonant points) and a double amplitude of 0.02 inches

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 10 hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10 hour period specified above may be composed of two 5 hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests

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by the Government inspector in accordance with the following: (Sampling tests shall be conducted only when Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational tests at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction , initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be recorded and analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781. For the Qualification Phase, Test Level B shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples for the production equipment shall be tested as outlined in MIL-STD-781 (as modified herein), under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase, Test Level B shall be used.

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4.4.3.2.1 Lot Size for Sampling Phase - The equipment constructed during one month shall be one lot. The Accept-Reject criteria for Test Plan IV shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract or order, no equipments shall be shipped until an Accept decision is reached under each lot.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possiblity of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 2000 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the

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following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 2000 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 10,000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ± 1.3 volts (at a frequency of 400 ± 5 Hz)

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

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4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production acceptance and life tests shall be reconditioned by the contractor by replacing all wear of damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The test set and applicable accessory cables shall utilize a rectified 615 volts d-c nominal output for hi-pot testing individual programmed circuits within various AMAC systems. It shall also be capable of utilizing the 28 volt d-c output of the AMAC system to perform continuity checks of programmed AMAC circuits. The test set shall be used in testing the following AMAC systems:

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(1) Aero 6B

(2) A/A24B-4(V)

(3) T-249A

(4) T-375

(5) T-414-101

(6) T-414-57

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (See 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) The Drawings of DL 67A47F1 - The drawings shall have precedence over other referenced specifications. Any deviation from the drawings shall be specifically approved by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Nonrepairable Subassemblies - As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is

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greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be mearer 50,000 hours.

Project No. 1190-N001

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Naval Air Systems Command
Washington, D.C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81802(AS) TEST SET, ARMAMENT WIRING AN/AWM-38

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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